

NICOLE VALENTINA SANTANA

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PROFESSIONAL PROFILE

Degree candidate in Data Science with strong training in machine learning, deep learning, and Bayesian statistics. Experience in predictive model design, data structures, programming, and advanced visualization for Big Data analysis, with proficiency in backend solution development, numerical methods, and optimization techniques. Applied experience with GenAI and electrophysiological signal processing (computational neuroscience). Rigorous analytical thinking, effective problem-solving for complex challenges, and clear technical communication. Adaptable and collaborative, with a constant focus on quality, ethics, and delivering results that generate measurable impact.

WORK EXPERIENCE

AI/ML Engineer Intern, IBM

Jan 2025 – Jul 2025

- Optimized backend solutions for financial applications focused on contract management and APIs, improving system efficiency and scalability
- Implemented automated processes for file analysis and decryption, ensuring dependency traceability and generating structured reports
- Conducted advanced analysis of transactional databases, identifying behavioral patterns and proposing data-driven solutions
- Designed dynamic dashboards in Power BI, Streamlit, and Cognos, transforming complex data into strategic insights for technical and managerial teams
- **Achievements:** Developed a GenAI-based automatic classification engine with 96% accuracy, reduced critical incidents through predictive analysis, and created an AIOps chatbot that decreased incident resolution time by 35%
- Collaborated with multidisciplinary teams to implement AI solutions in financial environments, impacting workflows used by more than 500 internal users

Data Analyst / BI Developer (Enterprise Project), Catch Consulting

Feb 2023 – Jun 2023

- Led the design and development of a Business Intelligence solution for the HR department of a client company in Mexico, transforming operational reports into executive dashboards for strategic decision-making
- Defined and implemented KPIs and segmentation models to analyze performance, turnover, and productivity, aligned with organizational goals and business metrics
- Applied ethical visualization principles and data storytelling to present findings to executive audiences and leadership teams
- Worked under a weekly deliverables framework, with stakeholder reviews and iterative adjustments based on client feedback
- **Achievement:** Winning project of the corporate data visualization competition, selected as the best solution among multiple participating teams

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL
- **Machine Learning and Deep Learning:** TensorFlow, PyTorch, scikit-learn
- **Data Visualization:** Power BI, Streamlit, Cognos, Matplotlib, Seaborn
- **Statistical Analysis:** NumPy, Pandas, SciPy
- **Big Data:** Apache Spark, Hadoop
- **Signal Processing:** OpenCV, signal processing libraries
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Version Control:** Git, GitHub
- **Development:** APIs, Backend Optimization

TECHNICAL AND RESEARCH PROJECTS

Researcher / Analyst, Universidad Externado de Colombia Aug 2024 – Nov 2025
Influence of Brain Waves on the Learning of Artificial Neural Networks

- Designed and compared bio-inspired architectures with standard deep learning models for image classification tasks
- Implemented experiments in Python using TensorFlow, PyTorch, NumPy, and Matplotlib
- Developed visualizations to interpret model behavior during training
- Applied statistical techniques to validate results and evaluate robustness against noise

Researcher / Analyst, Universidad Externado de Colombia Nov 2024 – Dec 2024
Impact of Time Pressure on Neurophysiological Responses

- Analyzed neurophysiological and behavioral data from 40 officers using EEG and GSR signals
- Processed and analyzed data to identify correlations between brain activity and moral decision-making
- Implemented quantitative statistical analysis using sociodemographic surveys
- Developed insights into neuroethics and decision-making in high-pressure environments

Researcher / Analyst, Universidad Externado de Colombia Aug 2023 – Nov 2023
Renewable Energy Prediction

- Implemented and compared multiple linear regression models using Bayesian and frequentist approaches
- Analyzed per capita renewable energy generation capacity using a global database
- Developed predictive models to evaluate sustainable energy policies

Researcher / Analyst, Universidad Externado de Colombia Feb 2023 – Apr 2023
Digital Image Manipulation Detection

- Developed a system to detect alterations in digital images using frequency-domain analysis
- Implemented Fourier Transform techniques to detect anomalous patterns and discontinuities
- Used Python, OpenCV, NumPy, and Matplotlib to extract spectral fingerprints
- Achieved high accuracy in detecting subtle manipulations through segmentation

EDUCATION

Bachelor's Degree in Data Science with Micro-Certification in Neuroscience, 2022 – 2025
Universidad Externado de Colombia

Open Year Program, Universidad de los Andes, Colombia Jan 2021 – Jun 2021

Academic High School Diploma, Colegio La Merced 2020

Technical High School Program: Basic Sciences Applied to Engineering, 2018 – 2020
Universidad Sergio Arboleda

ADDITIONAL TRAINING

- **Software Development From A to Z** – Udemy (2025)
- **Artificial Intelligence & Machine Learning for Business** – Udemy (2025)
- **Azure DevOps: Build and Deploy App Using CI/CD Pipeline** – Udemy (2025)
- **Monitoring with Prometheus, Grafana, Spring Boot, and Docker** – Udemy (2025)
- **DevOps Bootcamp** – IBM (2025)
- **Data Analysis and Visualization Bootcamp** – CYMETRIA (2024)
- **Image Processing Seminar – Seeing the Invisible** – Universidad Externado de Colombia (2023)
- **Code in Place Bootcamp** – Stanford University (2023)
- **Skills Development for Innovation in the Fourth Industrial Revolution** – Universidad Nacional de Colombia (2020)

LANGUAGES

- **English:** C2
- **French:** B1

HONORS & AWARDS

- **Best Presentation** – First Data Visualization Competition (Universidad Externado, Jun 2023)
- **Excellence Medal** – Colegio La Merced (Nov 2020) – Awarded for top academic performance of the graduating class

Notes

- For more information, visit my LinkedIn profile: [linkedin.com/in/nicolevalentina-santana](https://www.linkedin.com/in/nicolevalentina-santana)
- To view the Spanish version of this CV, click the following link: [Spanish CV](#)